

Solution Architecture

Date: 30 September 2026 | Team ID: 6ab2240ffc67a5bc0222524ef | Project: ComicCraft – AI Comic Story Creator

High-level solution architecture

Layer	Components	Responsibility
Presentation / Client	Browser, HTML, CSS, Jinja2 templates	Collect creative inputs and display the comic preview/export state.
Application	FastAPI, routes.py, schemas.py	Validate requests, orchestrate generation and expose HTML/JSON endpoints.
AI / Generation	Outline generator, story generator, Hugging Face image generation	Create structured story content and panel illustrations.
Composition	layout_builder.py	Match panel text and image assets into a consistent five-panel layout.
Export	exporters.py + fpdf2	Render each panel as a PDF page and save the final artifact.
Storage	static/panels, static/exports	Store generated panel images and exported PDF files.

Request path

Browser → /generate → PromptRequest validation → generate_outline() → generate_story() → generate_image() × 5 → build_comic_layout() → save_pdf() → comic_preview.html

Configuration path

Environment variables → app/config.py → model/API configuration → generation services. API keys should remain local and should not be committed.